

6. Development Plan

The project development will follow several phases.

Phase 1 — Project Setup

Tasks:

- create project structure
- initialize frontend and backend
- establish API communication

Phase 2 — Map Interface

Tasks:

- integrate Leaflet map
- display OpenStreetMap tiles
- implement start and destination selection

Phase 3 — Obstacle Integration

Tasks:

- query building data from Overpass API
- convert building polygons to obstacles
- implement manual obstacle drawing

Phase 4 — Route Planning

Tasks:

- implement A* algorithm
- convert obstacles into planning model

- generate routes

Phase 5 — Simulation

Tasks:

- animate drone movement
- display route progress
- display drone altitude

Phase 6 — Mission Commands

Tasks:

- implement return-home command
- implement landing command
- implement mission reset

Phase 7 — Testing and Finalization

Tasks:

- test mission workflow
- refine UI
- finalize documentation